Ronaldo Ross

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Education

Carleton University Ottawa, ON
Bachelor of Computer Science(BCS) - Algorithms, Minor in Mathematics *Sept 2025 – Jun 2029*

Technical Design Teams

Software Developer May 2026 – Present
BlackBird UAV (BBUAV) *Ottawa, ON*

- Engineered Python automation scripts deployed on Linux-based Raspberry Pi companion computers to orchestrate real-time mission logic and waypoint execution.
- Architected telemetry pipelines using MAVLink protocol to bridge Mission Planner with onboard Python runtimes, enabling dynamic autonomous command injection.
- Implemented Software-in-the-Loop (SITL) simulation workflows to validate script stability, memory usage, and execution permissions before airfield deployment.
- Hardened onboard Linux environments by configuring system properties and shell permissions, eliminating resource bottlenecks in high-latency telemetry streams.

Projects

Sentinel: ESP32 Fault Recovery Engine | *C, ESP-IDF, RTOS, BLE, HTTPS, Spring* July 2026 – Aug 2026

- Engineered a reusable **ESP32 fault-recovery engine** that detects boot loops and autonomously recovers bricked firmware through secure over-the-air updates without physical device intervention.
- Designed a **safe-mode recovery system** with NVS-backed boot/crash persistence, **defensive asserts**, custom flash partitions, core-dump diagnostics, BLE provisioning, and a CLI for device recovery and forced OTA updates.
- Built a custom **HTTPS OTA pipeline** with Spring Boot, dual OTA partitions, streamed firmware binaries, SHA-256 integrity verification, and autonomous recovery from crash-loop conditions.
- Developed a **Unity/CMock test suite** covering recovery logic, OTA behavior, failure paths, and device-state transitions, with **CI/CD** automation for repeatable firmware validation.

StreamForge | *C, ESP-IDF, FreeRTOS, DMA, SPI, Unity* Aug 2026 – Present

- Developed a **DMA-driven Quad-SPI streaming engine** on the ESP32-S3 utilizing FreeRTOS synchronization and **double buffering** to enable continuous 4KB concurrent data transfers with minimal CPU intervention.
- Designed reusable **buffer allocation and memory-management interfaces** across multiple function calls, handling pointer-to-pointer ownership and dynamically allocated streaming buffers.
- Implemented **unit and mocked hardware tests** with Unity and CMock to validate SPI/DMA control flow, buffer management, synchronization, and failure paths across the streaming pipeline.

ARES-32: Distributed IoT Telemetry & Backend Pipeline | *C, Java, Spring, Docker* June 2026 – July 2026

- Engineered an end-to-end telemetry system integrating an **ESP32-S3 (ESP-IDF)** node with a containerized backend, achieving sub-second latency for real-time monitoring.
- Developed a high-reliability firmware stack using **FreeRTOS**, implementing thread-safe IPC via **Queues and Mutexes** to manage concurrent **I2C** sensor-reads and **MQTT** publishing.
- Architected a robust data pipeline using **cJSON**, **MQTT**, and **PostgreSQL**, containerizing multi-service infrastructure with **Docker Compose** and environment-injected configuration.
- Implemented a **BLE Provisioning Interface** using **NimBLE** for secure, out-of-band network configuration and credential handoff.
- Achieved high system reliability by implementing a comprehensive test suite using **JUnit 5 and Mockito** for service logic and **Integration Testing** with an **H2 in-memory database** to validate E2E data integrity from MQTT ingestion to PostgreSQL.

Technical Skills

Languages: C, C++, Java (17/21), Python, SQL (PostgreSQL), Lua, HTML, CSS, JavaScript
Embedded & IoT: ESP-IDF, FreeRTOS (RTOS), NVS, NimBLE, I2C, SPI, UART, MQTT, OTA, DMA, Coredumps
Backend & Cloud: Spring Boot, REST APIs, Qt, FastMCP (MCP), cJSON, JDBC, Hibernate/JPA, JUnit/Mockito
Developer Tools: Docker, Docker Compose, Git, Linux, IntelliJ, VS Code, GDB
Architecture: Multi-threaded IPC, Data Structures & Algorithms, State Machines, Mutex/Semaphores, Memory Management, Defensive programming